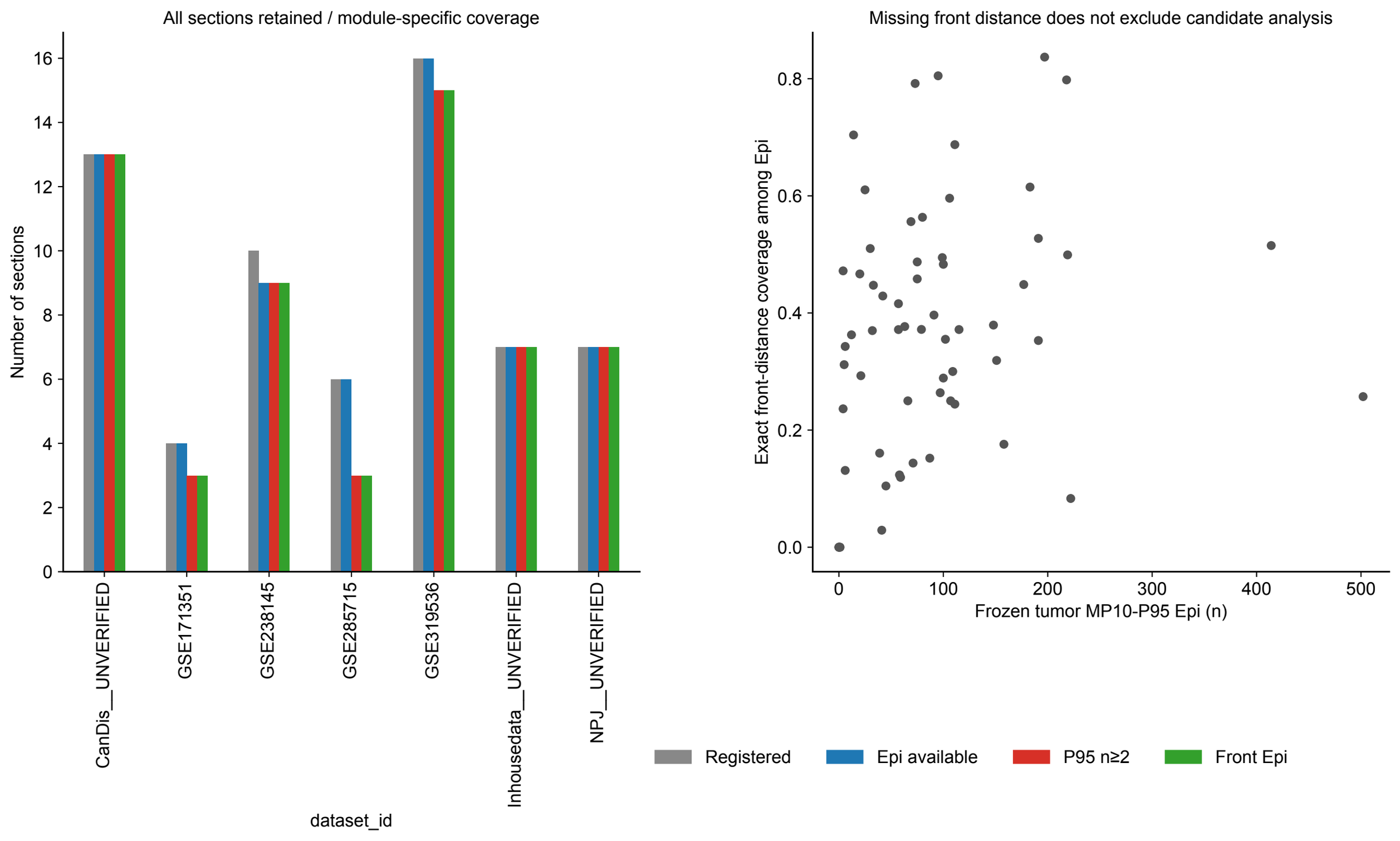
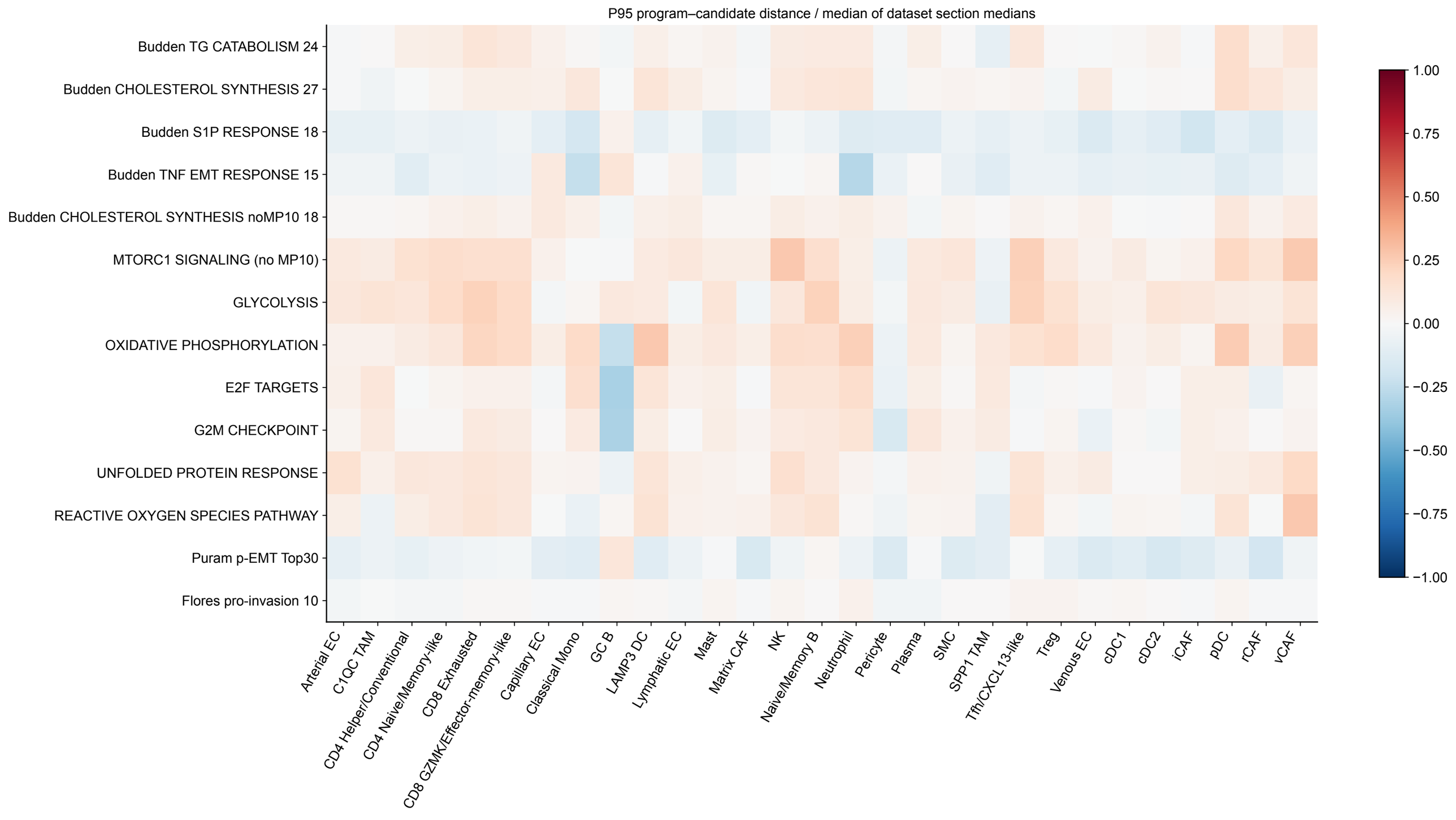
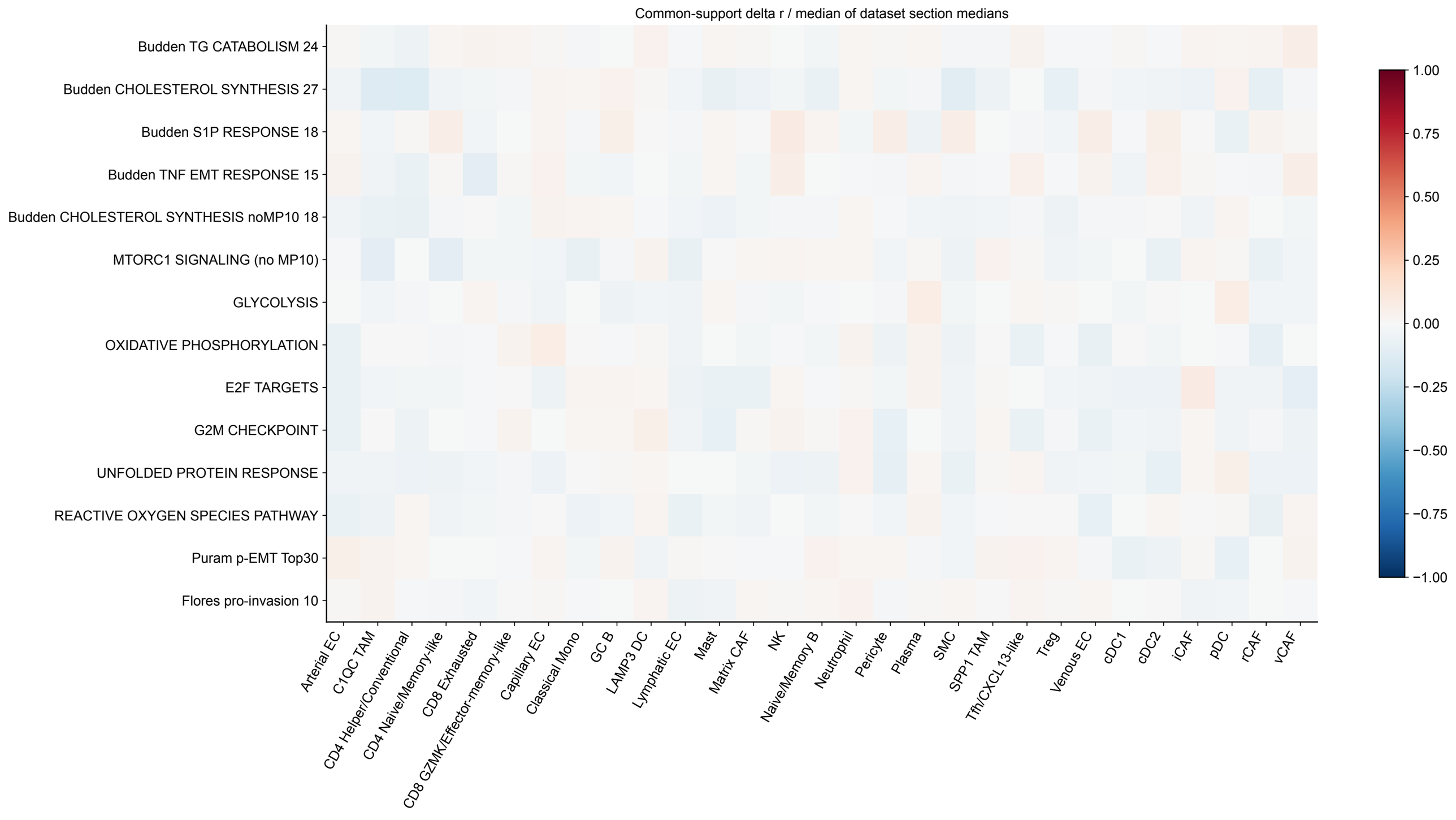

All sections retained / module-specific coverage
Number of sections
0
2
4
6
8
10
12
14
16
CanDis__UNVERIFIED
GSE171351
GSE238145
GSE285715
GSE319536
Inhousedata__UNVERIFIED
NPJ__UNVERIFIED
dataset_id
Registered
Epi available
P95 n≥2
Front Epi
Missing front distance does not exclude candidate analysis
Exact front-distance coverage among Epi
0.0
0.2
0.4
0.6
0.8
0
100
200
300
400
500
Frozen tumor MP10-P95 Epi (n)


P95 program–candidate distance / median of dataset section medians
Budden TG CATABOLISM 24
Budden CHOLESTEROL SYNTHESIS 27
Budden S1P RESPONSE 18
Budden TNF EMT RESPONSE 15
Budden CHOLESTEROL SYNTHESIS noMP10 18
MTORC1 SIGNALING (no MP10)
GLYCOLYSIS
OXIDATIVE PHOSPHORYLATION
E2F TARGETS
G2M CHECKPOINT
UNFOLDED PROTEIN RESPONSE
REACTIVE OXYGEN SPECIES PATHWAY
Puram p-EMT Top30
Flores pro-invasion 10
Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Common-support delta r / median of dataset section medians
Budden TG CATABOLISM 24
Budden CHOLESTEROL SYNTHESIS 27
Budden S1P RESPONSE 18
Budden TNF EMT RESPONSE 15
Budden CHOLESTEROL SYNTHESIS noMP10 18
MTORC1 SIGNALING (no MP10)
GLYCOLYSIS
OXIDATIVE PHOSPHORYLATION
E2F TARGETS
G2M CHECKPOINT
UNFOLDED PROTEIN RESPONSE
REACTIVE OXYGEN SPECIES PATHWAY
Puram p-EMT Top30
Flores pro-invasion 10
Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
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CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 raw Pearson / Budden CHOLESTEROL SYNTHESIS 27 / full ranges

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 raw Pearson / Budden CHOLESTEROL SYNTHESIS noMP10 18 / full ranges

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 raw Pearson / Budden S1P RESPONSE 18 / full ranges

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 raw Pearson / Budden TG CATABOLISM 24 / full ranges

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 raw Pearson / Budden TNF EMT RESPONSE 15 / full ranges

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

All frozen Epi: continuous MP10 AUCell vs candidate distance

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

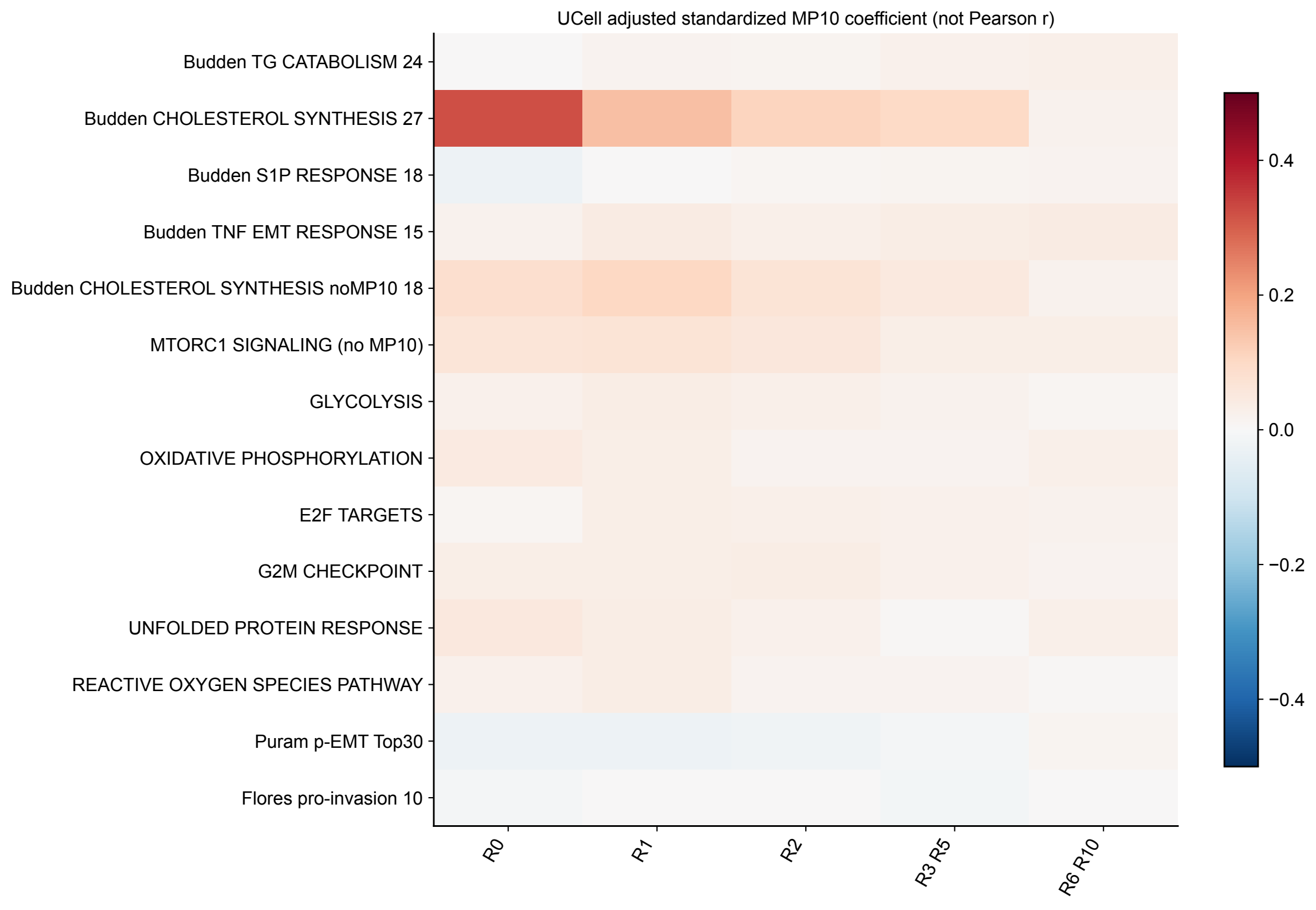
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

P95 minus nonP95 median candidate distance (µm; display clipped ±1000)

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

Arterial EC
C1QC TAM
CD4 Helper/Conventional
CD4 Naive/Memory-like
CD8 Exhausted
CD8 GZMK/Effector-memory-like
Capillary EC
Classical Mono
GC B
LAMP3 DC
Lymphatic EC
Mast
Matrix CAF
NK
Naive/Memory B
Neutrophil
Pericyte
Plasma
SMC
SPP1 TAM
Tfh/CXCL13-like
Treg
Venous EC
cDC1
cDC2
iCAF
pDC
rCAF
vCAF

1000
750
500
250
0
−250
−500
−750
−1000

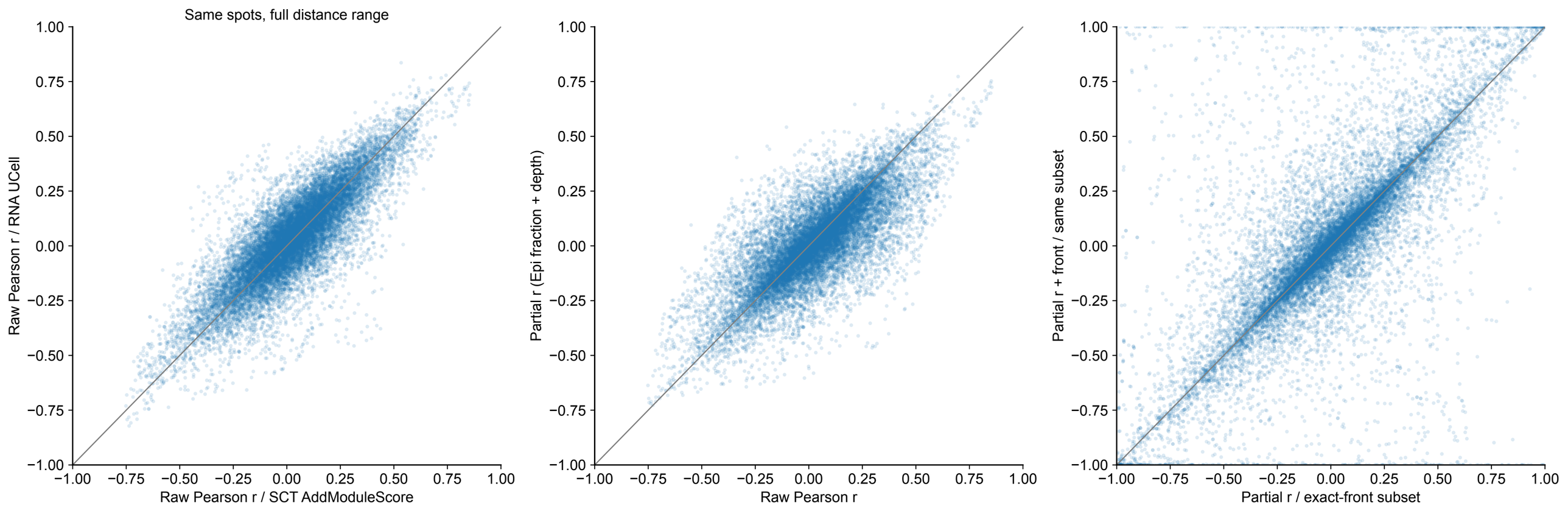

UCell adjusted standardized MP10 coefficient (not Pearson r)
Budden TG CATABOLISM 24
Budden CHOLESTEROL SYNTHESIS 27
Budden S1P RESPONSE 18
Budden TNF EMT RESPONSE 15
Budden CHOLESTEROL SYNTHESIS noMP10 18
MTORC1 SIGNALING (no MP10)
GLYCOLYSIS
OXIDATIVE PHOSPHORYLATION
E2F TARGETS
G2M CHECKPOINT
UNFOLDED PROTEIN RESPONSE
REACTIVE OXYGEN SPECIES PATHWAY
Puram p-EMT Top30
Flores pro-invasion 10
R0
R1
R2
R3 R5
R6 R10
0.4
0.2
0.0
−0.2
−0.4

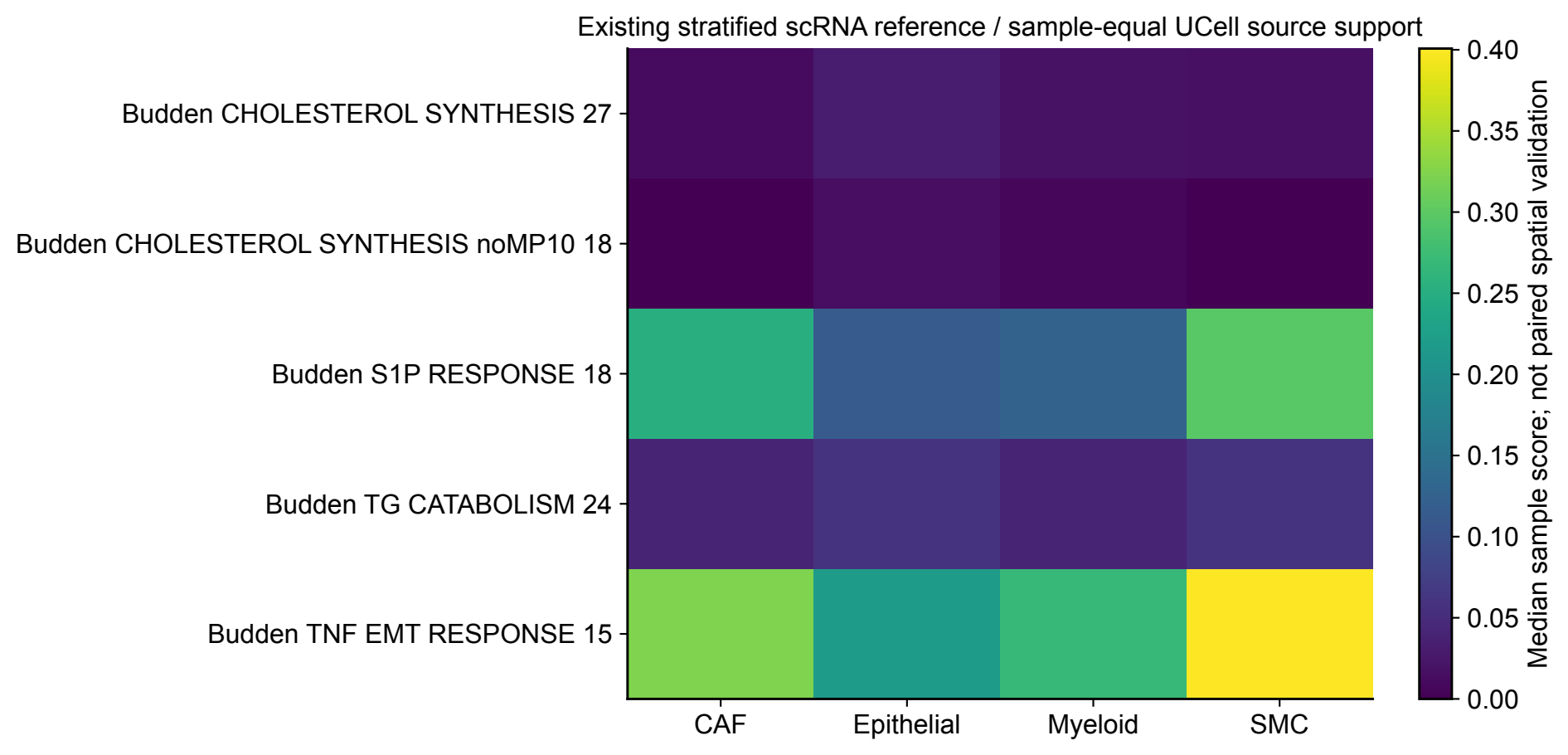
P95 minus nonP95 R12 proportion / front-only / percentage points

CanDis 2026 Sample1
CanDis 2026 Sample10
CanDis 2026 Sample11
CanDis 2026 Sample12
CanDis 2026 Sample13
CanDis 2026 Sample2
CanDis 2026 Sample3
CanDis 2026 Sample4
CanDis 2026 Sample5
CanDis 2026 Sample6
CanDis 2026 Sample7
CanDis 2026 Sample8
CanDis 2026 Sample9
GSE171351 A1
GSE171351 B1
GSE171351 C1
GSE171351 D1
GSE238145 B10
GSE238145 B11
GSE238145 B13
GSE238145 B14
GSE238145 B15
GSE238145 B16
GSE238145 B17
GSE238145 B18
GSE238145 B5
GSE238145 B9
GSE285715 N1
GSE285715 N2
GSE285715 N3
GSE285715 T1
GSE285715 T2
GSE285715 T3
GSE319536 BC01
GSE319536 BC02
GSE319536 BC04
GSE319536 BC05
GSE319536 BC08
GSE319536 BC10
GSE319536 BC11
GSE319536 BC12
GSE319536 BC13
GSE319536 BC14
GSE319536 BC15
GSE319536 BC16
GSE319536 BC17
GSE319536 BC18
GSE319536 BC19
GSE319536 BC21
Inhousedata T4384654
Inhousedata T4398343
Inhousedata T4413900
Inhousedata T4463209
Inhousedata T4465935
Inhousedata T4469479
Inhousedata T4471438
NPJ sample1
NPJ sample2
NPJ sample3
NPJ sample4
NPJ sample5
NPJ sample6
NPJ sample7

0 1 2 3 4 5 6 7 8

40
20
0
−20
−40


Same spots, full distance range
Raw Pearson r / RNA UCell
Raw Pearson r / SCT AddModuleScore
Partial r (Epi fraction + depth)
Raw Pearson r
Partial r + front / same subset
Partial r / exact-front subset
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Existing stratified scRNA reference / sample-equal UCell source support
Budden CHOLESTEROL SYNTHESIS 27
Budden CHOLESTEROL SYNTHESIS noMP10 18
Budden S1P RESPONSE 18
Budden TG CATABOLISM 24
Budden TNF EMT RESPONSE 15
CAF
Epithelial
Myeloid
SMC
Median sample score; not paired spatial validation
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00